

Math 548
Portfolio Assignment 3

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Problem 1 (W20, P1). Let A be a real $n \times n$ matrix and let A^T denote its transpose.

- (a) Prove that $\langle Av, w \rangle = \langle v, A^T w \rangle$ for all vectors $v, w \in \mathbf{R}^n$. *Hint:* Recall that the dot product $\langle u, v \rangle$ equals the matrix product $u^T v$.
- (b) Suppose now that A is also symmetric; i.e., that $A^T = A$. Also suppose v and w are eigenvectors of A with different eigenvalues. Prove that v and w are orthogonal.

Proof. (a) We have:

$$\begin{aligned}\langle Av, w \rangle &= (Av)^T w \\ &= (v^T A^T) w \\ &= v^T (A^T w) \\ &= \langle v, A^T w \rangle.\end{aligned}$$

(b) Let $Av = \lambda_1 v$ and $Aw = \lambda_2 w$. By the equation found in part (a), we have:

$$\begin{aligned}0 &= \langle Av, w \rangle - \langle v, A^T w \rangle \\ &= \langle \lambda_1 v, w \rangle - \langle v, Aw \rangle \\ &= \langle \lambda_1 v, w \rangle - \langle v, \lambda_2 w \rangle \\ &= \lambda_1 \langle v, w \rangle - \lambda_2 \langle v, w \rangle \\ &= (\lambda_1 - \lambda_2) \langle v, w \rangle.\end{aligned}$$

Since λ_1 and λ_2 are distinct eigenvalues, it must be the case that $\langle v, w \rangle = 0$. Thus v and w are orthogonal. $\square$

Problem 2 (F24, P4). Let $T : V \rightarrow V$ be a linear transformation on a finite-dimensional vector space. Prove that if $T^2 = T$, then:

$$V = \text{im}(T) \oplus \ker(T).$$

Proof. Let $x + y \in \text{im}(T) + \ker(T)$. Since $x \in \text{im}(T) \subseteq V$ and $y \in \ker(T) \subseteq V$, we have $x + y \in V$, which gives $\text{im}(T) + \ker(T) \subseteq V$. Now let $v \in V$. Note that:

$$\begin{aligned}T(v - T(v)) &= T(v) - T(T(v)) \\ &= T(v) - T(v) \\ &= 0.\end{aligned}$$

Hence $v - T(v) = k$, for some $k \in \ker(T)$. We can rewrite this as $v = k + T(v)$, where $k \in \ker(T)$ and $T(v) \in \text{im}(T)$. Thus $v \in \text{im}(T) + \ker(T)$, giving $V \subseteq \text{im}(T) + \ker(T)$.

Let $v \in \text{im}(T) \cap \ker(T)$. Then $v \in \text{im}(T)$ and $v \in \ker(T)$. Since $v \in \text{im}(T)$, there exists $w \in V$ such that $T(w) = v$. Since $v \in \ker(T)$, we have $T(v) = 0$. Together:

$$\begin{aligned} v &= T(w) \\ &= T(T(w)) \\ &= T(v) \\ &= 0. \end{aligned}$$

Since $\text{im}(T) \cap \ker(T) = \{0\}$ and $V = \text{im}(T) + \ker(T)$, we have $V = \text{im}(T) \oplus \ker(T)$. □

Problem 3 (S17, P2). Suppose F is a field and A is an $n \times n$ matrix over F . Suppose further that A possesses distinct eigenvalues λ_1 and λ_2 with $\dim \text{Null}(A - \lambda_1 I_n) = n - 1$. Prove A is diagonalizable.

Proof. Let $\{v_1, v_2, \dots, v_{n-1}\}$ be a basis for E_{λ_1} and let v_n be an eigenvector for E_{λ_2} . Suppose:

$$a_1 v_1 + \dots + a_{n-1} v_{n-1} + a_n v_n = 0 \tag{1}$$

Applying A to Equation (1) gives:

$$a_1 \lambda_1 v_1 + \dots + a_{n-1} \lambda_1 v_{n-1} + a_n \lambda_2 v_n = 0. \tag{2}$$

Multiplying both sides of Equation (1) by $-\lambda_2$ gives:

$$-a_1 \lambda_2 v_1 - \dots - a_{n-1} \lambda_2 v_{n-1} - a_n \lambda_2 v_n = 0 \tag{3}$$

Adding Equation (2) and Equation (3) yields:

$$(\lambda_1 - \lambda_2) a_1 v_1 + \dots + (\lambda_1 - \lambda_2) a_{n-1} v_{n-1} = 0.$$

Since $\lambda_1 \neq \lambda_2$, and since each $v_1, \dots, v_{n-1}$ is nonzero, it must be the case that $a_1 = a_2 = \dots = a_{n-1} = 0$. This implies that Equation (1) equals:

$$a_n v_n = 0.$$

Since v_n is an eigenvector, it is nonzero. Hence $a_n = 0$, establishing $\{v_1, \dots, v_{n-1}, v_n\}$ as a linearly independent set. Since $\dim(F^n) = n$ the linearly independent set of eigenvectors $\{v_1, \dots, v_{n-1}, v_n\}$ is in fact a basis for F^n . Since F^n admits a basis of eigenvectors, it is diagonalizable. □

Problem 4. Suppose $T : \mathbf{R}^n \rightarrow \mathbf{R}^n$ is a linear transformation with distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$, and let $v_1, v_2, \dots, v_m$ be corresponding eigenvectors. Prove $v_1, v_2, \dots, v_m$ are linearly independent.

Proof. Suppose towards contradiction $\{v_1, v_2, \dots, v_m\}$ is not linearly independent. Then there exists scalars $a_1, a_2, \dots, a_m$, not all of which are zero, such that $a_1v_1 + a_2v_2 + \dots + a_mv_m = 0$. After relabeling, choose k such that:

$$a_1v_1 + a_2v_2 + \dots + a_kv_k = 0, \quad (4)$$

where each $a_1, \dots, a_k$ is nonzero and k is minimal. Note that $2 \leq k \leq n$. Applying T to Equation (4) gives:

$$a_1\lambda_1v_1 + a_2\lambda_2v_2 + \dots + a_k\lambda_kv_k = 0. \quad (5)$$

Multiplying Equation (4) by $-\lambda_k$ gives:

$$-a_1\lambda_kv_1 - a_2\lambda_kv_2 - \dots - a_k\lambda_kv_k = 0. \quad (6)$$

Adding Equation (5) and (6) together yields:

$$(\lambda_1 - \lambda_k)a_1v_1 + (\lambda_2 - \lambda_k)a_2v_2 + \dots + (\lambda_{k-1} - \lambda_k)a_{k-1}v_{k-1} = 0.$$

Since each eigenvalue is distinct, $(\lambda_i - \lambda_k) \neq 0$ for each $1 \leq i \leq k-1$. Moreover, by construction each $a_1, \dots, a_k$ is nonzero. This contradicts the minimality of k , so it must be that the set $\{v_1, v_2, \dots, v_m\}$ is linearly independent. $\square$

Problem 5 (S18, P5). Suppose V is a finite-dimensional real vector space and $T : V \rightarrow V$ is a linear transformation. Prove that T has at most $\dim(\text{im } T)$ distinct nonzero eigenvalues.

Proof. Let $\dim(\text{im } T) = k$. Suppose towards contradiction there exists eigenvectors $v_1, \dots, v_{k+1}$ corresponding to distinct non-zero eigenvalues $\lambda_1, \dots, \lambda_{k+1}$. By Problem 4, the set $\{v_1, \dots, v_{k+1}\}$ is linearly independent. In particular, since each λ_i is nonzero, we have $T\left(\frac{v_i}{\lambda_i}\right) = v_i \in \text{im}(T)$. Thus $\{v_1, \dots, v_{k+1}\} \subseteq \text{im}(T)$ is a linearly independent set, which implies $\dim(\text{im } T) \geq k+1$. This is a contradiction, hence T must have at most $\dim(\text{im } T)$ distinct nonzero eigenvalues. $\square$